

Akash Karthik

San Jose, CA - akashkarthik473@gmail.com - linkedin.com/in/akash-karthik473 - github.com/akashkarthik473 - 9254069507

EDUCATION

Bachelor of Science, Computer Science

Expected May 2027

San Jose State University, San Jose, CA

Relevant Coursework: Data Structures & Algorithms, Linear Algebra, Discrete Math, Computer Systems, Intro to Circuits

TECHNICAL SKILLS

Languages: C, C++, Python, Assembly, Java

Embedded: Real-time firmware, bare-metal drivers, deterministic task scheduling, watchdogs, fixed-point PI/PID control

Hardware: STM32H755 (dual-core ARM Cortex-M7/M4), TTC 60 VCU, PX4, CAN (dual-bus, DBC), SPI, datasheet bring-up

Test & Validation: Software- and hardware-in-the-loop, regression suites, plant modeling, bus capture and trace analysis

Tools: TASKING, TTC Downloader, BusMaster, MoTeC i2 Pro, GDB, Git, CSV telemetry

EXPERIENCE

Software Lead

Jun 2026 - Present

Spartan Racing, Formula SAE | San Jose, CA

- Leading an in-house controller program replacing a vendor unit with custom STM32H755 hardware, running a 10 ms deterministic control task under a watchdog that reboots the controller if that task hangs
- Directing a seam-based port of a **5,400-line** production codebase, fixing the sensor and bus layer as the interface so everything below is rewritten for new silicon while control logic above carries over
- Built its CAN driver across both controller buses and specified, from the datasheet, the analog front-end changes needed before the board could be connected to hardware
- Leading a model-, software-, and hardware-in-the-loop effort so control code is validated against simulated plant models before reaching hardware, and set code standards across a modeling team of **7 senior engineers**

Software Engineer Intern

Feb 2026 - Present

Argus Defense | San Jose, CA

- Developed and validated embedded flight-control firmware in C on PX4, iterating control gains against logged flight telemetry
- Integrated and tuned visual-inertial odometry for onboard state estimation, improving position-hold accuracy
- Tuned PID flight-control loops to reduce vertical and horizontal position drift in autonomous flight

Software Controls Engineer

Jul 2025 - Jun 2026

Spartan Racing, Formula SAE | San Jose, CA

- Implemented fixed-point PI torque control with saturation, anti-windup, and slew limits for deterministic execution
- Developed embedded power-limiting firmware with a PI controller, maximizing actuator output under a hard **80 kW** limit
- Built CAN diagnostics firmware for live power and torque frames, status bits, and error counters for bus monitoring

Software Intern

Aug 2024 - Jun 2025

Spartan Racing, Formula SAE | San Jose, CA

- Built deterministic entry and exit state logic for power limiting from measured power, requested torque, and threshold flags
- **Results:** 1st in endurance at MIS EV 2025

PROJECT EXPERIENCE

SR-Wireless-CAN — Wireless Telemetry & Flashing Platform

2026

Spartan Racing, Formula SAE | San Jose, CA | Backend lead

Python, python-can, FastAPI, WebSockets, SQLite, Docker

- Proposed and implemented wireless firmware flashing for the TTC 60 controller, decoding an undocumented flashing sequence from CAN trace captures and reimplementing it in Python over the bus
- Developed wireless CAN capture and diagnostics tooling with DBC-driven signal decoding, replacing tethered bench monitoring during hardware testing

Embedded Driver Display — Real-Time Instrument Cluster

Feb 2026 - Present

Spartan Racing, Formula SAE | San Jose, CA

C, STM32H7, CAN, SPI

- Built a real-time CAN telemetry display for high-voltage, throttle, temperature, and power signals, driving an LCD over SPI
- Designed an energy bar and LED alert system giving the operator lap-by-lap feedback on energy budget